

When Engines Cannot Talk to Each Other: Federated Learning for Aircraft Health Monitoring

Background: Imagine you are the chief aviation engineer at a busy international airline. Every day, thousands of passengers trust you to keep their flight safe. Each aircraft engine is equipped with many sensors — measuring temperature, pressure, vibrations, fuel flow, fan speed and literally everything — generating millions of **time-series data** points for every flight. Your job is to predict: **when will this engine fail or need maintenance?** Answer too early, and you pull a healthy engine off the wing for unnecessary maintenance — costing tens of thousands of dollars. Answer too late, and an engine fails mid-flight, which can be more catastrophic. This prediction problem — estimating the **Remaining Useful Life (RUL)** of an engine — is one of the most important and most difficult challenges in modern aviation engineering. It sits at the heart of what researchers call Prognostics and Health Management (PHM).

The good news is that deep learning has made remarkable progress here [1] [2] [3] [4]. Researchers have trained neural networks on sensor time-series to predict RUL with impressive accuracy on standard benchmark datasets. The bad news is that these models need a lot of data — especially those data from engines that fail, which is rare by design. A single airline simply does not see enough failures and variety of data to train a reliable model on its own. The obvious fix is to combine data from multiple airlines. Ten airlines pooling their failure data would give ten times the learning signal, and the resulting model would generalize far better to unusual degradation patterns that any one airline rarely encounters. But here is where the story hits a wall: airlines will not share their engine data. Engine degradation records reveal which aircraft are approaching removal, which routes face near-term capacity risk, and what maintenance costs are coming — information that is commercially sensitive, legally protected, and, as a 2025 data breach at a major European carrier demonstrated, actively targeted by competitors and bad actors. Centralizing raw sensor data is simply not an option.

This is exactly the problem that Federated Learning (FL) was designed to solve. Instead of moving data to one central server, FL moves the model to the data. Each airline trains a copy of the shared model on its own local data for a few rounds, then sends only the updated model weights — never the raw sensor readings — to a coordination server. The server combines all the weights into one improved global model and sends it back. Repeat this cycle enough times and every airline end up with a model that has learned from all fleets, without any airline ever exposing its private data. Recent research has shown this approach works for aircraft engine prognostics [5] [6] [7]. Federated frameworks have been built that jointly predict early fault risk and RUL. When ten airlines collaborate this way, RUL accuracy improves for eight out of ten compared to each training in isolation, which is a compelling result. This is the **baseline** we are going to work with. Here we have few unsolved **research questions**:

Machine Learning Side:

- RQ1 (Heterogeneity & Personalization): Real-world airline fleets do not all use identical sensors. One operator might log 21 sensor channels (attributes), another only 15, and a third an entirely different subset. Basic federated PHM models assume every participant has the same inputs — which means in practice you either reduce everyone to the lowest common denominator or exclude some operators entirely. How can we map each operator's variable sensor inputs into a shared representation space so that no data needs to be dropped or any operator excluded? [8].
- RQ2 (Class Imbalance across Clients): Some airlines rarely experience engine failures due to better maintenance or newer fleets. As a result, they have very few failure examples. When their models are

combined with others that have more failure data, their learning signal becomes weak. This can cause poor fault detection performance for these data-scarce airlines. The question is how to design aggregation methods that protect weak failure signals instead of suppressing them.

- RQ3 (Root cause & interpretability): Engineers cannot rely only on a predicted RUL value. They need to understand why a prediction was made, which sensors are driving that prediction and what physical component those sensor patterns point to. How does linking sensor-level attribution (identifying which sensor readings contributed most to a given prediction) to an aviation maintenance ontology (a structured knowledge base mapping sensor patterns to known fault modes and components) change the interpretability and actionability of RUL predictions for ground engineers?
- RQ4 (Concept Drift): Current federated PHM models, including Paper [9], assume the sensor-to-RUL relationship is stationary throughout training. If a fault mode shift is injected mid-training on one client, how quickly does that client's AUPRC degrade, and can a lightweight drift-detection trigger on the local classifier head recover performance within 10 FL rounds?
- RQ5 (Validation under Non-IID): In paper [10], each airline tests other airlines' models on its own data and gives them a score — a low score means the model contributes less to the next global update. The problem is that a model trained on Arctic short-haul data will naturally score poorly when tested on tropical long-haul data, not because it is a bad model, but simply because the two operations look nothing alike. The scoring system punishes difference, not poor quality. How much does this distributional mismatch distort the aggregation weights, and how does accounting for it change per-operator RUL accuracy?

Security Side:

- RQ6 (Membership Inference): During federated training, each operator never shares raw sensor data — only the model weights updated from that data. But those weights are not as safe as they seem. An attacker sitting between an operator and the central server, simply capturing those weight updates, may be able to work backwards and recover sensitive information about that operator's private engine data — a well-known attack called membership inference. The question is how much sensitive information can be recovered and how to prevent such leakage.
- RQ7 (Model poisoning): A malicious airline operator could deliberately send corrupted weight updates to the server, pushing the global model to predict healthier-than-real RUL for a competitor's engine type. How would such an attack be carried out in practice, and what defenses can detect or mitigate it?

Tasks:

1. Task 1: Baseline implementation

Implement the federated learning baseline using one or more aviation-relevant datasets. Suitable options include C-MAPSS, PHM, N-CMAPSS, Turbofan Engine Degradation Dataset, or among others. Run the distributed learning (mentioned in background) experiments with between 3 and 6 airline operators acting as FL clients and a single central FL aggregation server. The model should be capable of predicting both RUL and early fault detection.

2. Task-2: Address One or More Research Questions

Pick one or more (as many as you want) research questions (RQ1–RQ7), then design and implement (code) suitable solution(s). Since the questions vary in difficulty, they will not carry equal weight in your performance evaluation. Therefore, selecting the right problem(s) is important to ensure

feasibility within your limited time while also maximizing your overall performance. This choice should reflect your ability to make sound, strategic decisions.

3. Task-3: Future Directions

Go beyond the questions listed above. Identify additional research gaps, recommend future directions, and potential solutions based on your own exploration of the problem space.

Instructions and Deliverables:

1. What to Submit:

- a. Upload all code (including result logs) to a GitHub repository and share the link.
- b. For your findings, methodology, pipeline, and results, you may use any format you prefer: a Word document, a slide deck, a video presentation, or something else entirely. The deliverable does not need to be formal; it just needs to be clear and understandable to someone reviewing your work.

2. A Note on Failed Attempts:

Any failed attempt is valuable research finding, it tells the community which directions are not worth pursuing and demonstrates genuine effort. It is completely acceptable if your output is imperfect or incomplete. We value your effort and honesty, so share whatever you have.

3. A Note on AI Assistance:

Using AI tools or agents is highly encouraged. However, you will remain responsible for everything your agent produces, both the good and the bad! Keep enough oversight over your pipeline so that you can adjust it to meet any change in research needs. Don't fall into the AI hallucination trap, models can confidently generate plausible sounding but completely wrong stuff, so always cross-check the outputs yourself before treating them as fact.

References

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